

Roshan Karthik

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SKILLS

- **Languages:** C++, Python, C, Java
- **Tools/Platforms:** Jupyter Notebook, Google Colab, GitHub.
- **Frameworks:** NumPy, Pandas, Scikit-Learn, Keras, PyTorch
- **Soft Skills:** Problem-Solving, Team Player, Project Management, Adaptability

PROJECTS

Car Damage Detection Model | [Github](#)

Nov' 25

- Developed an end-to-end car-damage detection system using PyTorch (Torchvision CNN models), improving classification accuracy to 92% on test images and optimizing preprocessing with Pillow, reducing image-loading latency by 35%.
- Engineered a scalable Fast API backend for real-time damage-assessment inference, achieving <150ms API response time through model optimization, batching, and efficient GPU utilization.
- Launched an interactive Streamlit dashboard enabling users to upload vehicle images and receive severity predictions, increasing user interaction efficiency by 50% and supporting images up to 5MB with automatic resizing.
- **Tech Stack:** Python, Fast API, Streamlit, PyTorch, TorchVision, Pillow

House Price Prediction Model | [Github](#)

Nov' 25

- Architected a housing-price prediction pipeline using a dataset of 12,968 properties — covering price range from ~US \$85,000 to ~US \$2,350,000 (average price ~US \$982,626).
- Implemented data preprocessing, feature encoding (categorical → One-Hot), scaling, and model training, achieving model evaluation with standard metrics (RMSE, MAE, R²) as outlined in documentation.
- Designed an interactive web-based interface integrated with the prediction pipeline to deliver real-time property-price estimates based on user inputs.
- **Tech Stack:** Python, Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn.

TRAINING

AlgoTutor (Edtech Company) | [Certificate](#)

Aug' 25

Mastering Data Structures and Algorithms

- Strengthened algorithmic thinking by working with arrays, linked lists, stacks, queues, trees, graphs and hashing to deepen problem-solving ability.
- Constructed efficient C++ implementations for sorting and searching algorithms, improving precise code Performance and reducing time complexity across projects.
- Validated concepts by solving structured problem sets, enhancing competitive programming proficiency and coding accuracy.

CERTIFICATES

- Machine Learning Specialization | [Coursera](#) Sep' 25
- ChatGPT-4 Prompt Engineering: ChatGPT, Generative AI & LLM | [Infosys](#) Aug' 25
- AI-Powered-NLP | [AlgoTutor](#) Aug' 25

ACHIEVEMENTS

- Completed a 100-day continuous coding streak and solved 300+ Data Structures & Algorithms problems.
- Ranked #204 in a university-wide problem-solving leaderboard.

EDUCATION

Lovely Professional University

Bachelor of Technology

Computer Science and Engineering; CGPA: 7.90

Phagwara, Punjab

Aug' 23 – Present

Narayana Junior College

Intermediate

PCM; Percentage: 98.1%

Hyderabad, Telangana

Jun' 20 – Mar' 22

Sri Chaitanya Techno School

Matriculation

Percentage: 98.6%

Hindupur, Andhra Pradesh

Apr' 19 – May' 20